

CHP 4: TEXT CLASSIFICATION & CLUSTERING

Text Classification Problem:

Introduction to Text Classification: Text classification is a fundamental task in Natural Language Processing (NLP) that involves categorizing text into predefined categories. This problem is highly relevant in various applications, such as spam detection, sentiment analysis, news categorization, and more. The main goal is to automatically assign labels to text based on its content.

Understanding the Basics: Text classification works by analyzing the words, phrases, or even the entire content of the text to determine which category it belongs to. For instance, in email spam detection, the classifier decides whether an email is "spam" or "not spam." In sentiment analysis, it might classify a product review as "positive," "negative," or "neutral."

Steps Involved in Text Classification:

1. Data Collection:

- The first step is to collect the data that needs to be classified. This could be emails, social media posts, product reviews, news articles, etc. The data is typically labeled, meaning that each text has a corresponding category.

2. Preprocessing the Text:

- Raw text data needs to be preprocessed before feeding it to the classifier. Preprocessing includes steps like:
 - Tokenization: Splitting the text into individual words or tokens.
 - Lowercasing: Converting all text to lowercase to maintain uniformity.
 - Removing Stopwords: Removing common words like "the," "is," "in," etc., that don't contribute much to the meaning.
 - Stemming or Lemmatization: Reducing words to their root form (e.g., "running" to "run").
 - Vectorization: Converting text into numerical form using techniques like Bag of Words (BoW), TF-IDF (Term Frequency-Inverse Document Frequency), or word embeddings.

3. Choosing a Model:

- After preprocessing, the text is ready to be classified. Various machine learning models can be used for text classification, such as:
 - Naive Bayes: A simple probabilistic model based on Bayes' theorem, often used for text classification due to its efficiency.

- Support Vector Machines (SVM): A powerful model that tries to find the optimal boundary between different categories.
- Deep Learning Models: Models like Convolutional Neural Networks (CNN) or Recurrent Neural Networks (RNN) can also be used for more complex text classification tasks.

4. Training the Model:

- The selected model is trained using the labeled dataset. The model learns to recognize patterns and associations between the text and its corresponding category.

5. Testing and Evaluation:

- After training, the model is tested on a separate set of data to evaluate its performance. Common evaluation metrics include accuracy, precision, recall, and F1-score.

6. Deployment:

- Once the model performs well, it can be deployed in a real-world application where it can classify new, unseen text.

Challenges in Text Classification:

- **Ambiguity:** The same word or phrase can have different meanings depending on the context, making classification difficult.
- **Large Vocabulary:** The vast number of words and phrases in natural language increases the complexity.
- **Data Imbalance:** Sometimes, certain categories have more examples than others, leading to biased models.
- **Feature Engineering:** Deciding which features (words, phrases) to use for classification can be challenging.

Applications of Text Classification:

- **Spam Detection:** Automatically filtering out spam emails.
- **Sentiment Analysis:** Determining the sentiment of reviews or social media posts.
- **Topic Categorization:** Classifying news articles or documents by topic (e.g., sports, politics, technology).
- **Language Detection:** Identifying the language in which a text is written.

Feature Selection

Introduction to Feature Selection: Feature selection is a process used in machine learning to select a subset of relevant features (variables, predictors) for model construction. The goal is to improve the model's performance by reducing the dimensionality of the data, thus minimizing overfitting, speeding up the training process, and enhancing the model's interpretability.

Steps in Feature Selection:

1. Understanding the Data:

- Begin by understanding the dataset, including the number of features and their types (categorical, numerical). Each feature may contribute differently to the outcome.

2. Importance of Feature Selection:

- High-dimensional datasets can have many irrelevant or redundant features that do not contribute to the predictive power of the model. Removing these irrelevant features can lead to simpler, faster, and more accurate models.

3. Techniques for Feature Selection:

- **Filter Methods:** These methods rank features based on statistical tests (like correlation, chi-square) before the model is trained. They are independent of the learning algorithm.
 - Examples: Correlation matrix, Chi-Square test, Information Gain.
- **Wrapper Methods:** These methods evaluate the model's performance with different subsets of features and select the one that produces the best result.
 - Examples: Forward selection, Backward elimination, Recursive Feature Elimination (RFE).
- **Embedded Methods:** These methods perform feature selection during the model training process. The learning algorithm itself decides which features contribute most to the accuracy.
 - Examples: LASSO (Least Absolute Shrinkage and Selection Operator), Ridge Regression.

4. Evaluation of Selected Features:

- After selecting features, the model is trained and evaluated on a validation set. The performance is compared to the model trained on the full feature set. Metrics like accuracy, precision, recall, and F1-score are used for evaluation.

Challenges in Feature Selection:

- **Overfitting:** Selecting too many features may lead to a model that performs well on training data but poorly on unseen data.
- **Computational Cost:** Evaluating all possible subsets of features can be computationally expensive.
- **Domain Knowledge:** Sometimes, domain expertise is required to decide which features are relevant.

Applications of Feature Selection:

- **Improved Model Performance:** By reducing noise in the data, the model becomes more accurate.
- **Faster Computation:** Fewer features mean faster computation times, which is crucial for real-time applications.
- **Enhanced Interpretability:** With fewer features, it's easier to understand the model's decisions.

Conclusion: Feature selection is a critical step in the machine learning pipeline that enhances model performance and efficiency. By carefully choosing the most relevant features, we can build models that are not only accurate but also faster and easier to interpret.

Naive Bayes Text Classification with Numerical Example

Introduction to Naive Bayes: Naive Bayes is a probabilistic classifier based on Bayes' theorem. It's particularly popular for text classification tasks because it's simple, fast, and works well with large datasets.

Bayes' Theorem Recap:

The diagram illustrates Bayes' Theorem with the following components and annotations:

- LIKELIHOOD** (orange text): the probability of "B" being TRUE given that "A" is TRUE. An arrow points from this text to the $P(B|A)$ term in the numerator.
- PRIOR** (green text): the probability of "A" being TRUE. An arrow points from this text to the $P(A)$ term in the numerator.
- POSTERIOR** (green text): the probability of "A" being TRUE given that "B" is TRUE. An arrow points from this text to the $P(A|B)$ term on the left side of the equation.
- The probability of "B" being TRUE** (pink text): An arrow points from this text to the $P(B)$ term in the denominator.

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Numerical Example: Let's consider a simple example where we want to classify a message as either "spam" or "not spam."

Naive Bayes Classifier: A Numerical Example

Problem: Classify a message as either "spam" or "not spam."

Training Data:

Class	Message
Spam	Buy cheap pills
Spam	Cheap watches for sale
Spam	Get cheap loans
Not Spam	Meeting tomorrow
Not Spam	Lunch at 12

Not Spam Project deadline extended

Step 1: Calculate Priors

- $P(\text{Spam}) = 3/6 = 0.5$
- $P(\text{Not Spam}) = 3/6 = 0.5$

Step 2: Calculate Likelihood

- **For "cheap pills for sale" given Spam:**
 - $P(\text{"cheap"} \mid \text{Spam}) = 3/3 = 1.0$
 - $P(\text{"pills"} \mid \text{Spam}) = 1/3 = 0.33$
 - $P(\text{"for"} \mid \text{Spam}) = 2/3 = 0.67$
 - $P(\text{"sale"} \mid \text{Spam}) = 1/3 = 0.33$
- **For "cheap pills for sale" given Not Spam:**
 - $P(\text{"cheap"} \mid \text{Not Spam}) = 0/3 = 0.0$
 - $P(\text{"pills"} \mid \text{Not Spam}) = 0/3 = 0.0$
 - $P(\text{"for"} \mid \text{Not Spam}) = 0/3 = 0.0$
 - $P(\text{"sale"} \mid \text{Not Spam}) = 0/3 = 0.0$

Step 3: Calculate Posterior

- **For Spam:**
 - $P(\text{Spam} \mid \text{"cheap pills for sale"}) = 1.0 * 0.33 * 0.67 * 0.33 * 0.5 = 0.0367$
- **For Not Spam:**
 - $P(\text{Not Spam} \mid \text{"cheap pills for sale"}) = 0.0 * 0.0 * 0.0 * 0.0 * 0.5 = 0.0$

Conclusion: Since $P(\text{Spam} \mid \text{"cheap pills for sale"}) > P(\text{Not Spam} \mid \text{"cheap pills for sale"})$, the message is classified as "Spam."

k-Nearest Neighbors (k-NN) with Numerical Example

Introduction to k-NN: k-NN is a simple, instance-based learning algorithm that classifies a data point based on the majority class among its nearest neighbors.

Numerical Example: Suppose we have a dataset with two classes (A and B) and we want to classify a new point PPP.

Training Data:

- Class A: (1, 1), (2, 2)
- Class B: (4, 4), (5, 5)
- New Point $P=(3,3)$

Step 1: Choose k: Let's choose $k=3$.

Step 2: Calculate Distances:

Distance from P to (1, 1):

$$\begin{aligned}d &= \sqrt{(3-1)^2 + (3-1)^2} \\&= \sqrt{2^2 + 2^2} \\&= \sqrt{4 + 4} \\&= \sqrt{8} \\&\approx 2.83\end{aligned}$$

Distance from P to (2, 2):

$$\begin{aligned}d &= \sqrt{(3-2)^2 + (3-2)^2} \\&= \sqrt{1^2 + 1^2} \\&= \sqrt{1 + 1} \\&= \sqrt{2} \\&\approx 1.41\end{aligned}$$

Distance from P to (4, 4):

$$\begin{aligned}d &= \sqrt{(3-4)^2 + (3-4)^2} \\&= \sqrt{(-1)^2 + (-1)^2} \\&= \sqrt{1 + 1} \\&= \sqrt{2} \\&\approx 1.41\end{aligned}$$

Distance from P to (5, 5):

$$\begin{aligned}d &= \sqrt{(3-5)^2 + (3-5)^2} \\&= \sqrt{(-2)^2 + (-2)^2} \\&= \sqrt{4 + 4} \\&= \sqrt{8}\end{aligned}$$

≈ 2.83

Step 3: Determine Nearest Neighbors: The 3 nearest neighbors are:

- (2, 2) from Class A
- (4, 4) from Class B
- (1, 1) from Class A

Step 4: Majority Voting:

- Class A: 2 neighbors
- Class B: 1 neighbor

Conclusion: The new point PPP is classified as Class A.

**FOR KNN , K -MEANS , SVM AND HIERARCHICAL CLUSTERING
YOU CAN REFER ML CHP 3-7 AND 9**



